

Empowering global disease surveillance with CURED: a tool for rapid identification of unique genomic biomarkers

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ABSTRACT Rapid tracking of emerging pathogenic microorganisms is crucial for designing effective treatment, infection control, and prevention strategies. While whole-genome sequencing (WGS) offers the necessary granularity to track emerging clones, it remains prohibitively expensive at the scales needed to monitor with high resolution in real time. We present CURED (Classification Using Restriction Enzyme Diagnostics), which uses a training set of sequenced genomes to identify unique k-mers with restriction sites specific to a clonal lineage. CURED enables fast and inexpensive PCR-based diagnostic tests for surveillance or outbreak investigations with minimal use of WGS. Benchmarking against existing tools, CURED compares favorably and scales more efficiently than other k-mer search strategies. We validated and tested CURED in five distinct data sets: (i) previously identified biomarkers described for a methicillin-resistant *Staphylococcus aureus* (MRSA) clone in Rio de Janeiro, (ii) diagnostic alleles for different lineages in the USA300 MRSA clone, (iii) the extensively drug-resistant *Acinetobacter baumannii* Global Clone 1 lineage, (iv) toxigenic versus non-toxigenic *Clostridioides difficile*, and (v) circulating *S. aureus* clones in a neonatal intensive care unit (NICU). We implemented CURED as part of NICU infection prevention efforts and report the test's speed, sensitivity, and specificity in a real-world setting. CURED is a scalable, multithreaded, memory-, and cost-efficient pipeline tailored for rapid clone detection and restriction site analysis. While particularly impactful for localized outbreak investigations and targeted surveillance, our preliminary work at the global scale suggests broader implementation is feasible. CURED is freely available at <https://github.com/microbialARC/CURED>.

IMPORTANCE Timely and cost-effective detection of emerging microbial clones is essential for infection prevention and public health surveillance. While whole-genome sequencing remains the gold standard for tracking microbial evolution and transmission, its cost, infrastructure requirements, and turnaround time limit its scalability, especially in resource-limited settings. CURED addresses this gap by enabling the development of inexpensive, PCR-based diagnostic assays informed by genomic data, without requiring further sequencing. By identifying lineage-specific restriction sites through a scalable and memory-efficient k-mer pipeline, CURED enables the translation of genome-scale insights into actionable diagnostics. This tool supports broader implementation of genomic-informed diagnostics in both local and global pathogen surveillance efforts.

KEYWORDS clonal lineage tracking, rapid pathogen surveillance, microbial diagnostics, k-mer analysis, restriction enzyme PCR, hospital, infection prevention and control, outbreak, resource-limited settings

Editor Juliette Hayer, Institut de Recherche pour le Développement Delegation Regionale Occitanie, Montpellier, France

Peer Reviewer Jonathan C. Thomas, Nottingham Trent University, Nottingham, United Kingdom

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The authors declare no conflict of interest.

See the funding table on p. 13.

Received 17 July 2025

Accepted 22 January 2026

Published 19 March 2026

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Rapid dissemination of *Staphylococcus aureus* in the neonatal intensive care unit is associated with invasive infection

Received: 20 June 2025

Accepted: 23 January 2026

Published online: 09 February 2026

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Staphylococcus aureus is a leading cause of severe infections in neonatal intensive care units (NICUs). We present results of precision surveillance of *S. aureus* carriage and invasive infection in a cohort study of high-risk infants in the NICU with the goal of identifying novel strategies for prevention and control. Using whole genome sequence guided epidemiology over a 3-year timeframe, we identify spatial and temporal links for transmission between babies, suggesting shared spaces, caregivers, and physical proximity as major risks for transmission in the NICU. Moreover, environmental surveillance reveals potential environmental reservoirs of *S. aureus*. Remarkably, specific clusters of *S. aureus* strains that are associated with invasive infection are also detected in more infants over time during routine surveillance, suggesting a strong link between the rate of dissemination and disease in this vulnerable population. Overall, our findings demonstrate a strong association amongst colonization, transmission, persistence, and the development of invasive infections, underscoring the importance of targeted measures to prevent *S. aureus* infections in the NICU setting.

Infants hospitalized in Neonatal Intensive Care Units (NICUs) are at increased risk for healthcare-associated infections (HAIs), which are a major cause of morbidity, mortality, adverse long-term neurodevelopmental outcomes, and a significant contributor to the escalating cost of neonatal care^{1–3}. *Staphylococcus aureus* colonizes and persists

on the skin and mucous membranes of infants early in life and is a major cause of invasive infection^{4–6}. A recent study of 100,000 very preterm infants born across the US between 2018 and 2020 revealed that one-quarter of HAIs were attributed to *S. aureus*⁷. Single and multicenter investigations of *S. aureus* infections in NICUs have

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Using Genomic and Traditional Epidemiologic Approaches to Define Complex Transmission Pathways of *Klebsiella pneumoniae* Infection in a Neonatal Unit in Botswana, 2022–2023

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Abstract

Background: *Klebsiella pneumoniae* (*Kpn*) is a major cause of infant mortality worldwide, with most transmission occurring among hospitalized neonates in low- and middle-income countries where infections caused by multidrug-resistant *Kpn* (MDR-*Kpn*) are increasingly common. We hypothesized that integrating laboratory surveillance for neonatal colonization and infection, real-time epidemiologic investigations, and whole-genome sequencing (WGS) could identify transmission pathways to guide targeted infection prevention and control (IPC) strategies.

Methods and Findings

We conducted *Kpn* surveillance in a 36-bed neonatal unit in Botswana over 12 months (2022–2023). WGS was performed on *Kpn* isolates from bloodstream infections (BSIs), and MDR-*Kpn* isolates collected from environmental sampling during outbreaks and twice-monthly colonization screenings (skin and perirectal swabs) using culture media selective for MDR-*Kpn* (CHROMagar Extended-spectrum beta-lactamase [ESBL]/SuperCarba). WGS data were analyzed using multilocus sequence typing (MLST), pangenome and reference-based single-nucleotide polymorphism (SNP) analyses, and Bayesian phylogenetics.

We identified 55 *Kpn* BSIs during the 12-month surveillance period and the median prevalence of MDR-*Kpn* colonization was 28%. *Kpn* was recovered from multi-use intravenous (IV) fluid bags during a *Kpn* outbreak (41 BSIs, 10 deaths), which was controlled by implementing a 24-hour discard policy for IV medications. Among 270 *Kpn* isolates available (28 BSI, 232 colonizing, 10 environmental [six IV fluid, four sink drain]), WGS confirmed over half of BSI genomes (n=17) were ST1414, a clone susceptible to third-generation cephalosporins not detected during MDR-*Kpn* colonization screening, but closely related (<25 SNPs) to six *Kpn* isolates from contaminated IV fluids.

Conclusions

This study reinforces the value of integrating WGS with real-time epidemiologic investigations to understand transmission dynamics and guide IPC. Colonization surveillance focused solely on MDR-*Kpn* may overlook drug-susceptible but outbreak-prone strains.